

Trigonometry
Most repetitive questions
(Of all SSC Exams Last year)

Reciprocal Conjugates

$\sec\theta + \tan\theta = \dfrac{1}{\sec\theta - \tan\theta}$

$\operatorname{cosec}\theta + \cot\theta = \dfrac{1}{\operatorname{cosec}\theta - \cot\theta}$

Note: If $\sec\theta - \tan\theta > \sec\theta + \tan\theta$, then $\tan\theta < 0$ i.e. negative.

If $\operatorname{cosec}\theta - \cot\theta > \operatorname{cosec}\theta + \cot\theta$, then $\cot\theta < 0$ i.e. negative.

$\sec+\tan\theta = \frac{2}{5}$ $\quad a+b=3$ $\quad a-b=9$

$\sec-\tan\theta = \frac{5}{2}$ $\quad 6$ $\quad \frac{3-9}{2} = -3$ $\quad -ve$

Note: $1 \pm \sin\theta$ and $1 \pm \cos\theta$ are often used in questions.

$$\frac{1 \pm \sin\theta}{\cos\theta} = \sec\theta \pm \tan\theta \quad \text{and} \quad \frac{1 \pm \cos\theta}{\sin\theta} = \operatorname{cosec}\theta \pm \cot\theta$$

1 If secA + tanA = x, express secA in terms of x.

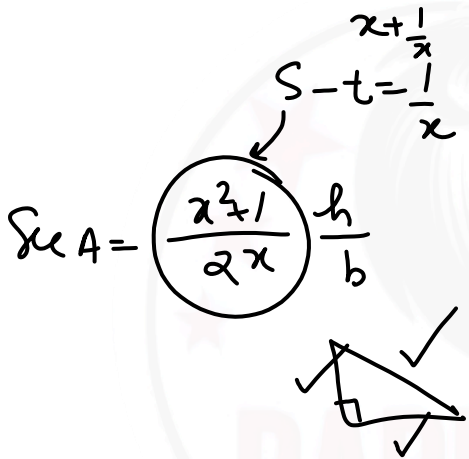
अगर secA + tanA = x, तो secA को x के पदों में व्यक्त कीजिए।

a) $(x^2 + 1)/2x$

b) $(x^2 + 1)/x$

c) 2

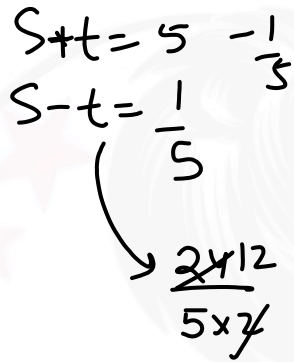
d) j



2 If tanA + secA = 5 and A is acute, then find the value of tanA.

अगर tanA + secA = 5 तथा A न्यूनकोण है, तो tanA का मान ज्ञात कीजिए।

a) 3
b) 4
c) 1
d) 2



3 If $secA + tanA = p$, then what is secA in terms of p?

यदि $secA + tanA = p$, तो p के संदर्भ में secA क्या है?

a) $\frac{p^2+1}{2p}$

b) $\frac{p^2+1}{3p}$

c) $\frac{p^2+1}{5p}$

d) $\frac{p^2+1}{9p}$

$s - t = \frac{1}{p}$

$\frac{p + \frac{1}{p}}{2}$

PARMAR SSC

4 If secθ + tanθ = 5, and θ lies in the first quadrant, find the value of sinθ.

यदि secθ + tanθ = 5 है, और θ प्रथम चतुर्थांश में स्थित है, तो sinθ का मान ज्ञात कीजिए।

a) 5/13
b) 12/13
c) 3/5
d) 4/5

$s+t = 5 + \frac{1}{5}$

$s-t = \frac{1}{5}$

$\sec\theta = \frac{13}{12}$ h

b

12 p

PARMAR SSC

$a+b=\checkmark\ 2(a^2+b^2)$

$a-b=\ 4ab$

$s-t=\frac{1}{\sqrt{3}}$

$s+t=\sqrt{3}$

$3-\frac{1}{3}$

$\frac{8}{3}$ ~~$\times st=$~~ ~~2~~

$\sqrt{3}+\frac{1}{\sqrt{3}}$

$\sec=\frac{\frac{4}{\sqrt{3}}}{2}=\frac{4}{\sqrt{3}\times 2}\ \frac{h}{b}$

1 P

S If secx − tanx = 1/√3, then the value of secx × tanx is:

यदि secx − tanx = 1/√3 हो, तो secx × tanx का मान क्या है?

~~a) 2/3~~

b) 4/√3

c) 2/√3

d) 4/3

$\frac{2}{\sqrt{3}}\times\frac{1}{\sqrt{3}}=\frac{2}{3}$

Complementary Angles

Sum $90°$

$\sin(90° - \theta) = \cos\theta$

$\tan(90° - \theta) = \cot\theta$

$\sec(90° - \theta) = \operatorname{cosec}\theta$

$\tan\theta = \frac{1}{\cot\theta}$

Question Types

$x + y = 90°$

Type (i)

$\sin x = \cos y$

$\tan x = \cot y$

$\sec x = \operatorname{cosec} y$

Example:

$\sin(2x + 15°) = \cos(3x - 10°)$, $x = ?$

Type (ii)

$\tan x \cdot \tan y = 1$

$\cot x \cdot \cot y = 1$

Example:

$\tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \cdots \tan 89^\circ = ?$

Type (iii)

$\sin^2 x + \sin^2 y = 1$

$\cos^2 x + \cos^2 y = 1$

Example:

$\sin^2 1^\circ + \sin^2 2^\circ + \cdots + \sin^2 89^\circ = ?$

Type (iv)

$\sec^2 x - \cot^2 y = 1$

$\operatorname{cosec}^2 x - \tan^2 y = 1$

Example:

$\sec^2 1^\circ + \sec^2 2^\circ + \ldots + \sec^2 45^\circ - \cot^2 46^\circ - \cot^2 47^\circ - \ldots - \cot^2 89^\circ = ?$

6 If $\sin(90° - x) = \cos(2x - 30°)$, and x is acute, find the value of x.

यदि $\sin(90° - x) = \cos(2x - 30°)$, तथा x न्यून कोण है, तो x का मान ज्ञात कीजिए।

a) 30°
b) 45°
c) 60°
d) 15°

$x + 60 = 90$

30

PARMAR SSC

7 If sin(90° − x) = cos(2x), then what is x?
यदि sin(90° − x) = cos(2x) है, तो x क्या है?

a) 0°
b) 30°
c) 25°
d) 15°

$x + 90 = 90$

PARMAR SSC

If tan A = cot(2A − 30°), then what is the value of A?

यदि tan A = cot(2A − 30°), तो A का मान क्या है?

a) 30°

b) 25°

c) 18°

d) 40°

$3A - 30 = 90$

PARMAR SSC

9 If $\sin x = \cos(5x - 60°)$, then what is the value of x?

यदि $\sin x = \cos(5x - 60°)$ है, तो x का मान क्या है?

a) 30°

b) 25°

c) 18°

d) 20°

$6x - 60 = 90$

150

PARMAR SSC

10 If $\sin(3x + 10°) = \cos(2x - 5°)$, find x.

यदि $\sin(3x + 10°) = \cos(2x - 5°)$ हो, तो x ज्ञात कीजिए।

a) 27°

b) 30°

c) 25°

d) 17°

$5x + 15 = 90$ 85

PARMAR SSC

|| If $\sin(2x + 15°) = \cos(x - 5°)$, what is the value of x?

HW

यदि $\sin(2x + 15°) = \cos(x - 5°)$, तो x का मान क्या है?

a) 45.57°

b) 60°

c) 30°

d) 26.67°

12 If $\sin(2x + 50)^\circ = \cos(2y - 50)^\circ$, such that $0^\circ < x + y < 90^\circ$, then find the value of tan(x + y).

यदि $\sin(2x + 50)^\circ = \cos(2y - 50)^\circ$ इस प्रकार है कि $0^\circ < x + y < 90^\circ$ है, तो $\tan(x + y)$ का मान ज्ञात कीजिए।

a) $\sqrt{2}$

b) $3\sqrt{3}$

c) 1

d) 0

$2x+2y = 90$

$x+y = 45$

PARMAR SSC

HW

13 If $\tan(x + 10°) = \cot(3x - 20°)$, what is the value of x?

यदि $\tan(x + 10°) = \cot(3x - 20°)$, तो x का मान क्या है?

a) $15°$

b) $25°$

c) $30°$

d) $20°$

SSC CHSL T1, 30 Nov 2025 S3

14 If $\tan(A) = \cot(B)$, what is the relationship between angle A and angle B?

यदि tan(A) = cot(B), तो कोण A और कोण B के बीच क्या संबंध है?

a) $A + B = 90°$

b) $A - B = 90°$

c) $A + B = 180°$

d) $A = B$

PARMAR SSC

15 If $\sin(90° - x) = \cos(2x)$, then what is x?
यदि $\sin(90° - x) = \cos(2x)$ है, तो x क्या है?

HW

a) 0°
b) 30°
c) 25°
d) 15°

PARMAR SSC

16 If sinx = cosy and cosx = siny, then x + y equals:
यदि sinx = cosy और cosx = siny, तो x + y बराबर है:

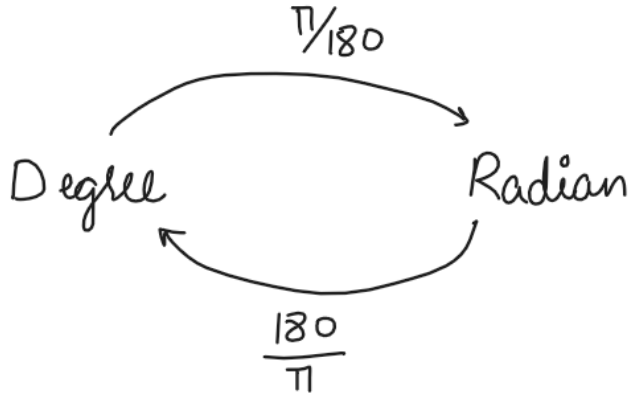
a) 45°
b) 60°
c) 90°
d) 180°

x+y=90 x+y=90

PARMAR SSC

Degree vs Radian

π radian $= 180^\circ$



$\pi^c = 180^\circ$

PAKMAK SSC

17 What is $\frac{5\pi}{4}$ radians in degrees?

$\frac{5\pi}{4}$ रेडियन कितने डिग्री में है?

a) 225°

b) 180°

c) 240°

d) 360°

$\frac{5\pi}{4} \times \frac{180}{\pi}$

$\frac{5 \times 180}{4}$ = 45

PARMAR SSC

17 What is the radian measure of 150°?

150° का रेडियन माप क्या है?

a) $\frac{5\pi}{6}$

b) $\frac{2\pi}{3}$

c) $\frac{3\pi}{4}$

d) $\frac{\pi}{3}$

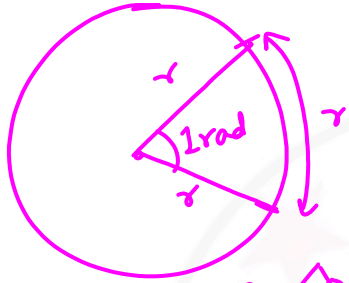
$150 \times \frac{\pi}{180} = \frac{5}{6}$

19 Convert $\frac{2\pi}{3}$ radians to degrees.

परिवर्तित करें $\frac{2\pi}{3}$ रेडियन को डिग्री में।

a) 120°
b) 90°
c) 180°
d) 60°

PARMAR SSC





Q. Convert 1.5 radians to degrees.

1.5 रेडियन को डिग्री में बदलो।

a) 90.56°

b) 120°

c) 85.94°

d) 180°

$\pi = \frac{22}{7}$

$\frac{3}{2} \times \frac{180 \times 7}{22} = \frac{945}{11} = 85\frac{10}{11}$

57°
285

2| Convert 3° to radians.

3° को रेडियन में बदलें।

a) $\frac{5\pi}{180}$

b) $\frac{5\pi}{360}$

c) $\frac{\pi}{60}$

d) $\frac{\pi}{180}$

$\frac{3\times\pi}{180}$

$\frac{\pi}{60}$

PARMAR SSC

Some Frequently Used Equations:

When $\theta = 45°$

$\frac{1}{\sqrt{2}} \times \sqrt{2}$

$\sin\theta + \cos\theta = \sqrt{2}$

$\sin\theta - \cos\theta = 0$ $\sin = \cos$

$\tan\theta + \cot\theta = 2$

$\tan\theta - \cot\theta = 0$

$\sec\theta + \operatorname{cosec}\theta = 2\sqrt{2}$

$\sec\theta - \operatorname{cosec}\theta = 0$

PARMAR SSC

22 If $tanA + cotA = 2$ and $A \in (0, \pi/2)$, then what is sinA + cosA?

यदि $tanA + cotA = 2$ और $A \in (0, \pi/2)$, तो sinA + cosA क्या है?

a) 1

b) $\sqrt{2}$

c) $\frac{1}{\sqrt{2}}$

d) Cannot be determined

23 If $\sin x = \cos x$, find $\sin^4 x + \cos^4 x$.

यदि $\sin x = \cos x$, तो $\sin^4 x + \cos^4 x$ ज्ञात कीजिए।

a) 1

b) $\frac{3}{4}$

c) $\frac{1}{2}$

d) $\frac{1}{4}$

$\frac{1}{4} \times 2$

PARMAR SSC

24 If $sinx + cosx = \sqrt{2}$, what is the value of sinx-cosx ?

यदि $sinx + cosx = \sqrt{2}, sinx - cosx$ का मान क्या है?

a) 0
b) 1
c) 2
d) 3

PARMAR SSC

25. If $sin\,\theta = cos\,\theta$, what is the value of θ (in degrees)?
अगर $sin\,\theta = cos\,\theta$, का मूल्य क्या है θ (in degrees)?

a) 45°
b) 35°
c) 30°
d) 40°

PARMAR SSC

26 If $\tan\theta + \cot\theta = 2$ and θ is an acute angle, find the value of $\tan^3\theta + \cot^3\theta$.

यदि $\tan\theta + \cot\theta = 2$ और θ एक न्यून कोण है, तो $\tan^3\theta + \cot^3\theta$ का मान ज्ञात कीजिए।

A. 0

B. 1

C. 2

D. 3

PARMAR SSC

27 If $sinA + cosA = 1$, then find the value of $\sin^4 A + \cos^4 A$.

यदि $sinA + cosA = 1, \sin^4 A + \cos^4 A$ का मान ज्ञात कीजिए।

a) 1

b) $\frac{1}{2}$

c) $\frac{3}{4}$

d) $\frac{5}{8}$

28 If $tan\theta + cot\theta = 5$, and θ is an acute angle, find the value of $\tan^2\theta + \cot^2\theta$.

यदि $tan\theta + cot\theta = 5$ है, और θ एक न्यून कोण है, तो $\tan^2\theta + \cot^2\theta$ का मान ज्ञात कीजिए।

a) 20
b) 21
c) 22
d) 23

sin3A → sinA

3A=θ

29 If $\sin 3A = \frac{3}{5}$, what *is* $\cos 3A$?

यदि $\sin \sin 3A = \frac{3}{5}$ है, तो $\cos 3A$ क्या है?

a) 4/5
b) 5/6
c) 7/8
d) 8/9

$\sin\theta = \frac{3}{5}$ P h b $\cos\theta = ?$

PARMAR SSC

30 If $\sin A = \frac{3}{5}$ and $\cos B = \frac{4}{5}$, and both A, B in first quadrant, then $\sin(A + B) =$?

यदि $\sin A = \frac{3}{5}$ और $\cos B = \frac{4}{5}$, और A, B दोनों प्रथम चतुर्थांश में हैं, तो $\sin(A + B) =$?

a) 12/25
b) 7/10
c) 24/25
d) 4/5

$\sin A \cos B + \cos A \cdot \sin B$

$\frac{3}{5} \times \frac{4}{5} + \frac{4}{5} \times \frac{3}{5} = \frac{24}{25}$

PARMAR
SSC